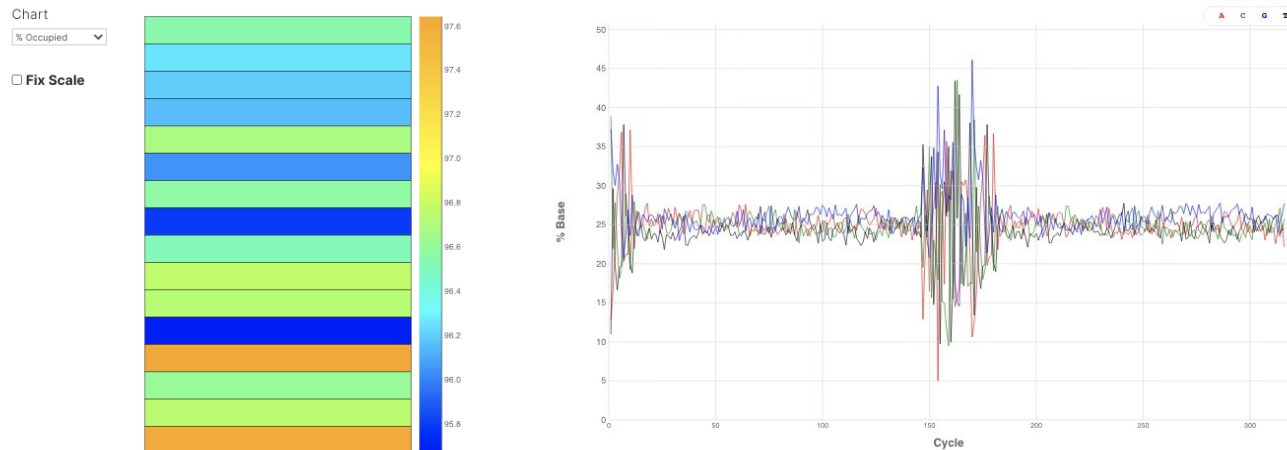


Sequencing run QC



Sequencing Run QC

Flow Cell: BPL20321-2824 Extracted: 318 Called: 318 Scored: 318

Flow Cell Chart

Chart

Intensity

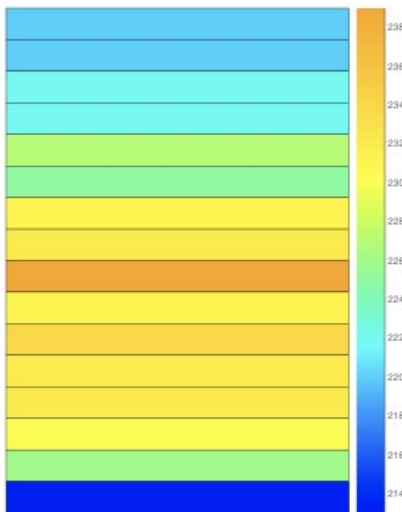
Cycle

Cycle 36

Channel

1

☐ Fix Scale



Data By Cycle

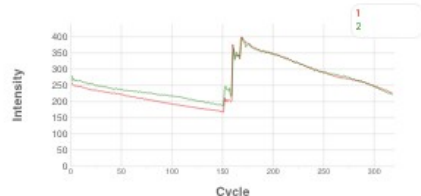
Chart

Intensity

Channel

All Channels

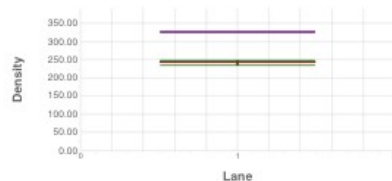
☐ Fix Scale



Data By Lane

Show

Density



QScore Distribution

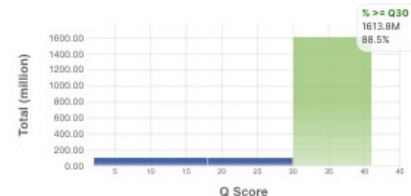
Read

All Reads

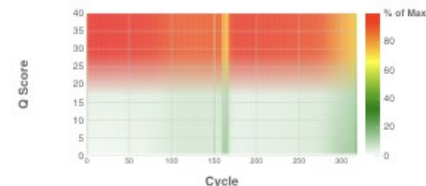
Cycle

All Cycles

☐ Fix Scale



QScore Heatmap



Did you get the expected amount and quality of sequencing data?

If not, why?

Sequencing Run QC using BaseSpace

After logging into BaseSpace, Click on 'Runs' to view a list of all your runs.
Click on a Run Name to see it's metrics.

illumina SEQUENCE HUB Manu Vanaersch...  

 HOME **RUNS** PROJECTS ANALYSES BIOSAMPLES APPS DEMO DATA   

Runs

ACTIVE PLANNED [NEW RUN](#) 

There are no urgent actions. Well done. Return in a few hours to check incoming runs.

 1 - 11 of 11 Show **25** 

 STATUS 	RUN NAME	AVG%Q30	%PF	INSTRUMENT 	CREATED
 Complete	COVID-19 ARTIC_Batch-10	 88.69%	74.62%	FS10000518	1/28/2021, 06:08:40

Sequencing Run QC using BaseSpace

COVID-19 ARTIC_Batch-10

SUMMARY BIOSAMPLES CHARTS METRICS INDEXING QC SAMPLE SHEET FILES



Instrument
FS10000518



Run Status
Complete

Lane QC Status
QcPassed



Flow Cell Status
QcPassed

Latest Analysis
--

Cycles
151 | 8 | 8 | 151

Yield
1.80 Gbp

Created
2021-01-28 06:40

Instrument Type
iSeq100

File Count/Size
[11,353 files](#) (1 GB)

File Status
Active

Owner

User

Flow Cell ID
BPL20321-2824

Run ID
20210128_FS10000...

AVG%Q30 and %PF



- **AVG%Q30** = percentage of reads that have bases with an average quality score >Q30
 - Ideally AVG%Q30 >90%, but >70% is good too
 - If <70%: evaluate why the quality is lower than expected. Possible reasons:
 - Overclustering: too much material was loaded on the flow cell
 - Adapter dimers: these will yield reads with only ~70bp of good signal, and the rest low quality 'nothingness'.
- **%PF** = percentage of passed filter reads = reads based on a cluster of a single molecule
 - iSeq: Ideally >70%, but >50% is good too
 - MiSeq/MiniSeq: Ideally >80%, but >70% is okay too
 - If <50%/<70%, check for indications of overclustering or underclustering

QC charts

Run: COVID-19 ARTIC_Batch-10: Charts

SUMMARY BIOSAMPLES **CHARTS** METRICS INDEXING QC SAMPLE SHEET FILES



Flow Cell: BPL20321-2824 Extracted: 318 Called: 318 Scored: 318

Flow Cell Chart

Chart

Intensity

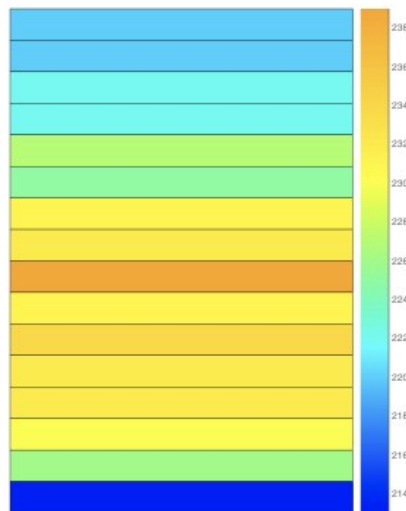
Cycle

Cycle 36

Channel

1

☐ Fix Scale



Data By Cycle

Chart

Intensity

Channel

All Channels

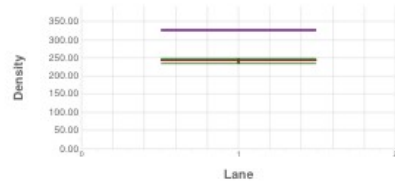
☐ Fix Scale



Data By Lane

Show

Density



QScore Distribution

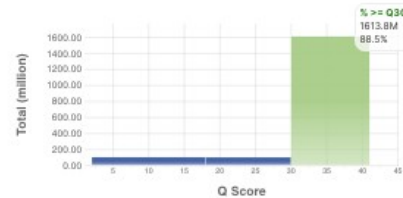
Read

All Reads

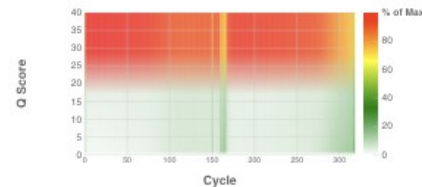
Cycle

All Cycles

☐ Fix Scale

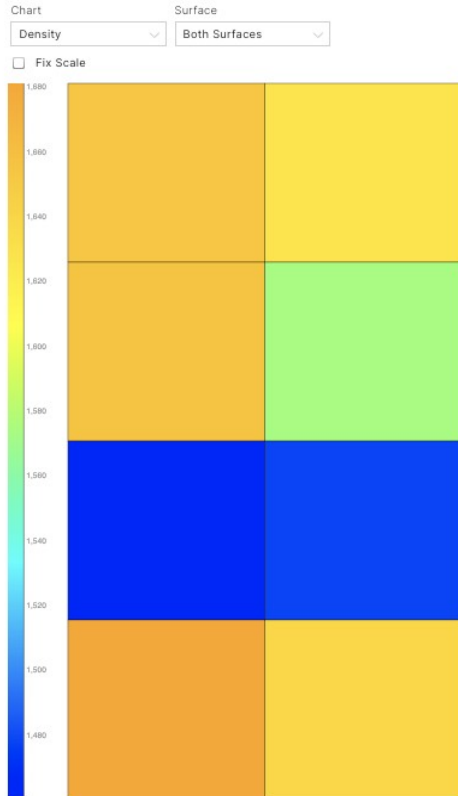


QScore Heatmap



Evaluate clustering for MiSeq

MiSeq



Flow Cell Chart: select **Density** under 'Chart'

Optimal Raw cluster density MiSeq:

- V2 kits: 1000-1200 K/mm²
- V3 kits: 1200-1400 K/mm²

This V3 example shows a raw cluster density ~1600 K/mm²

-> **The run was overclustered!**

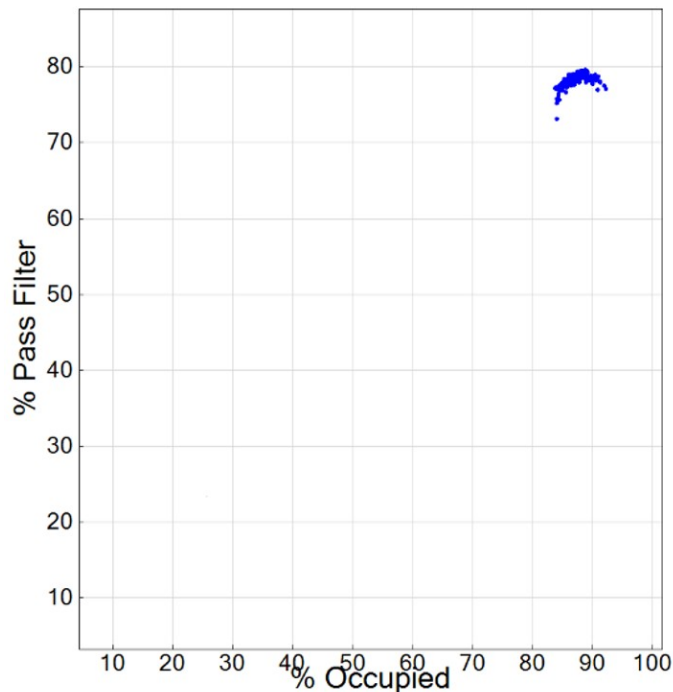
-> lower %PF reads

-> lower yield of PF reads total

Check QC data to see why concentration in nM of final library was underestimated (length overestimated?)



Evaluate clustering for iSeq/NextSeq and other patterned flow cells



In **SAV (not Basespace)**:

1. Load the run metrics data into SAV and then select the **Images** tab.
2. In the Images tab, select **Custom Plots**
3. In the Custom Plots window, ensure that the **Scatter plot** radio button is selected.
4. In the X Axis Value dropdown menu, select **%Occupied**
5. In the Y Axis Value dropdown menu, select **% Pass Filter**

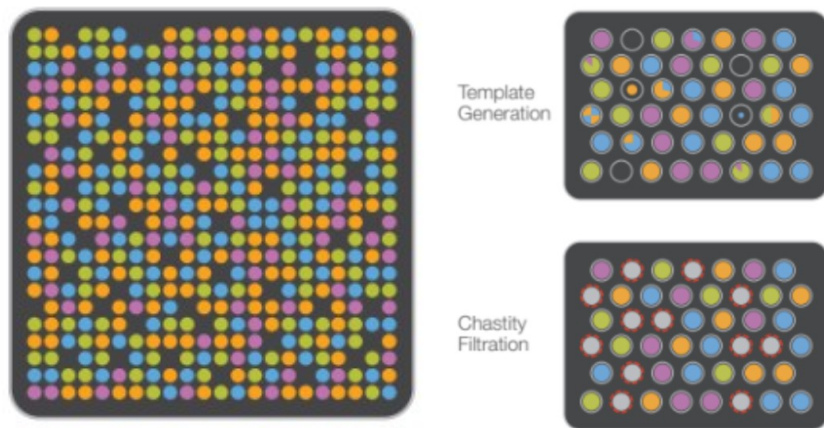


How is % PF calculated?

For patterned flow cells

All nanowells, **including empty wells** or clusters that may be dim, low quality, or polyclonal are included in the raw cluster count (denominator), which leads to lower %PF values.

$$\% \text{ PF} = \frac{\text{Clusters passing chastity filter}}{\text{Raw cluster count (aka \# of spaces in the nanowell)}}$$



A. Patterned

What could cause low %PF?

- Clusters are too close to their neighbors
- Clusters are empty

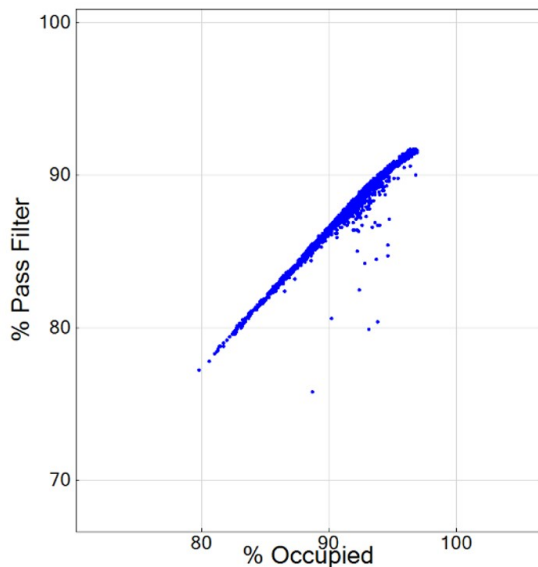
Evaluate clustering with % PF x % Occupied plots

For patterned flow cells

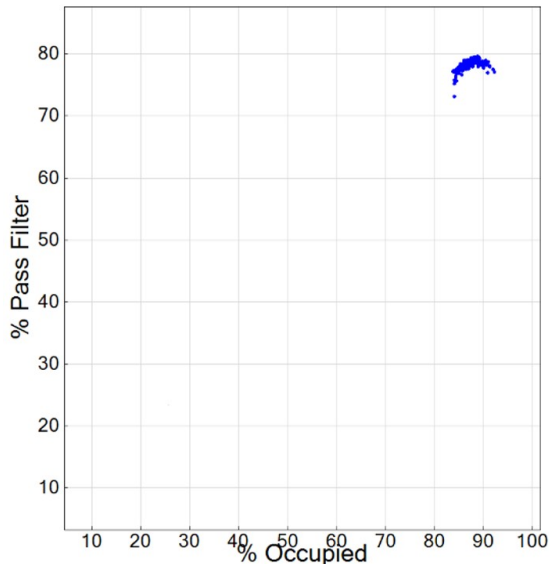
iSeq
NextSeq



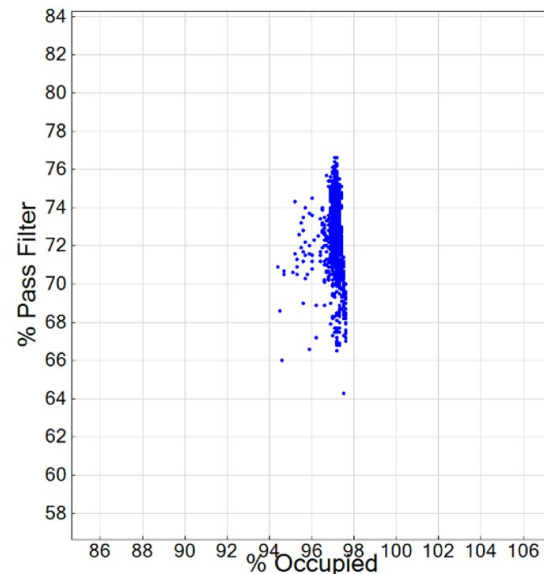
Under-clustered



Optimally clustered



Over-clustered



Evaluate clustering for iSeq/NextSeq – putting it together

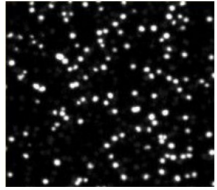
For patterned flow cells

	%PF	% Occupancy	Q30	PhiX alignment
Over-loaded	Low (clusters bleed into each other)	High	Low	Lower than expected*
Under-loaded	Low (many clusters are empty)	Low	High	Higher than expected*

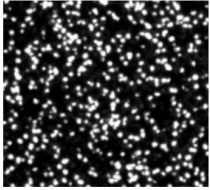
* PhiX alignment expected is based on the assumption that the PhiX library size (520bp) is ~the same as your library size

* PhiX must be uncompromised (freeze-thaws, DNase contamination, storage concentration)

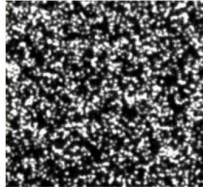
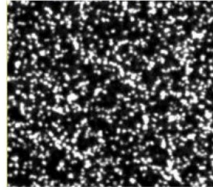
Evaluate clustering by looking at Files/Thumbnail_Images



Underclustered

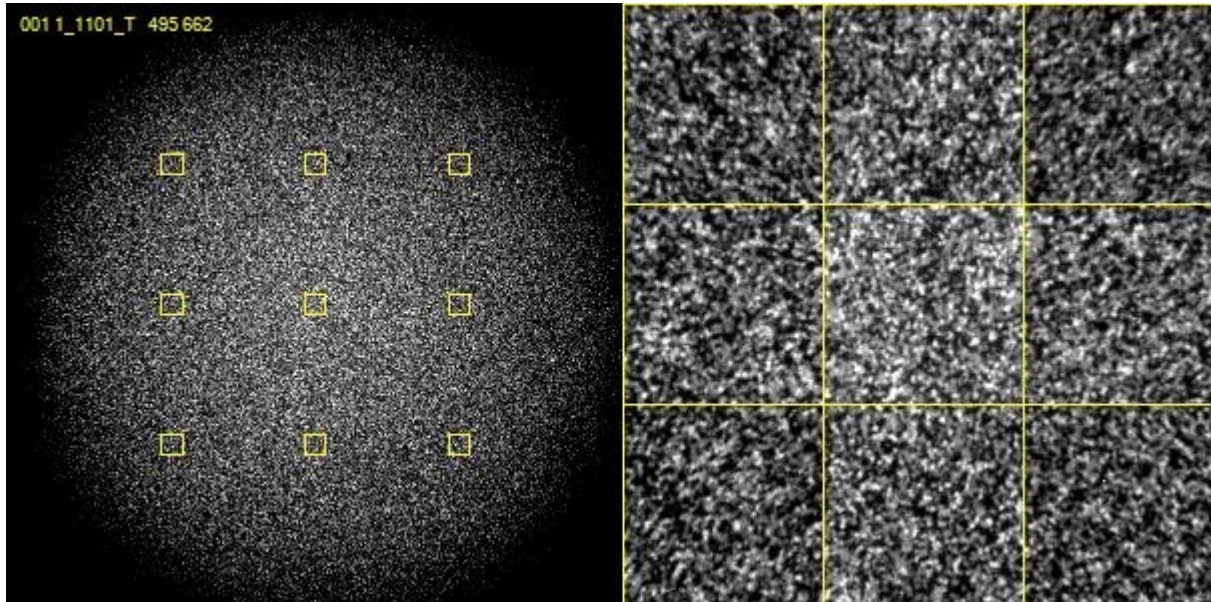


Optimal Clustering



Overclustered

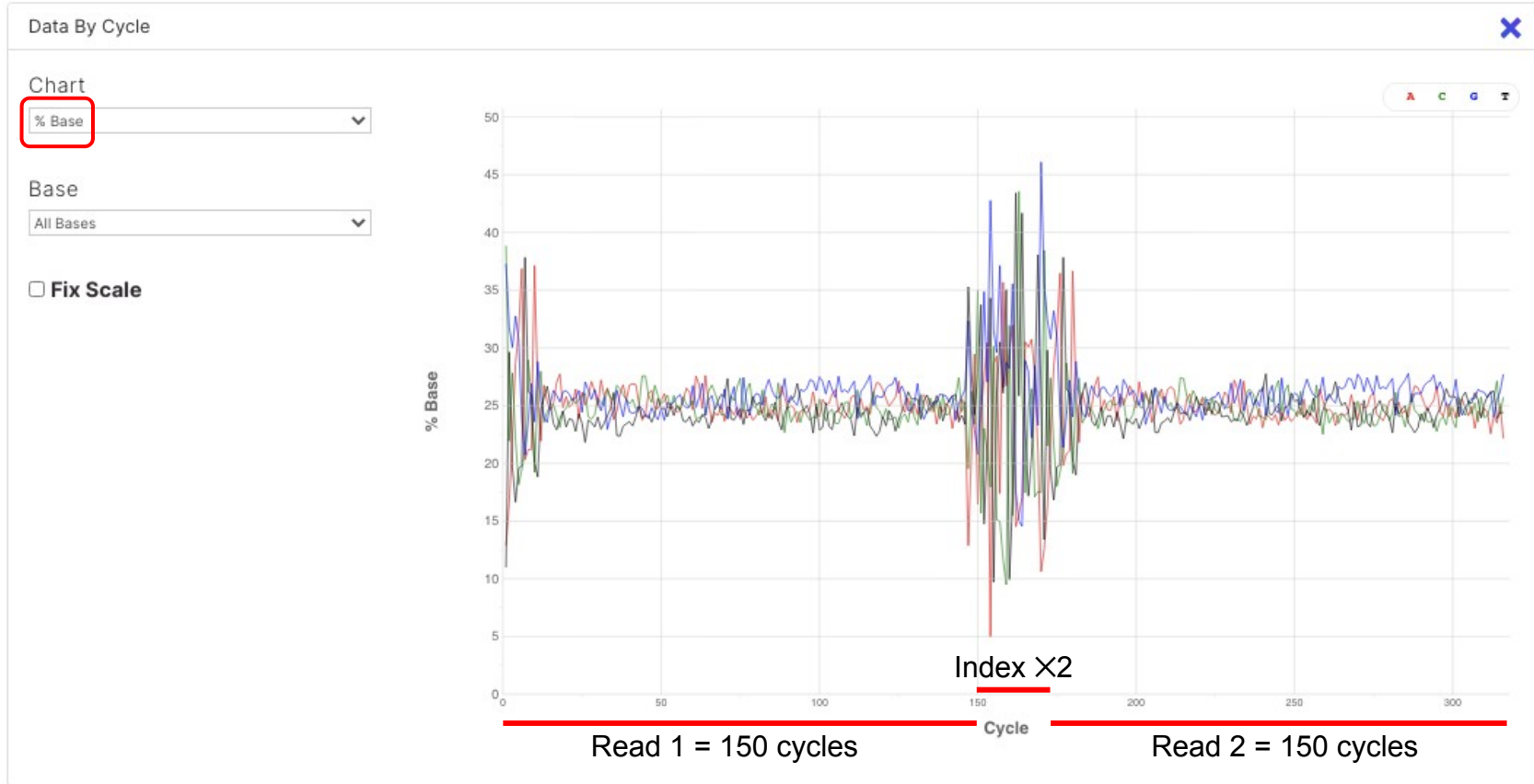
Illumina examples
from [here](#)



Tiles also show
overclustering of
this MiSeq run

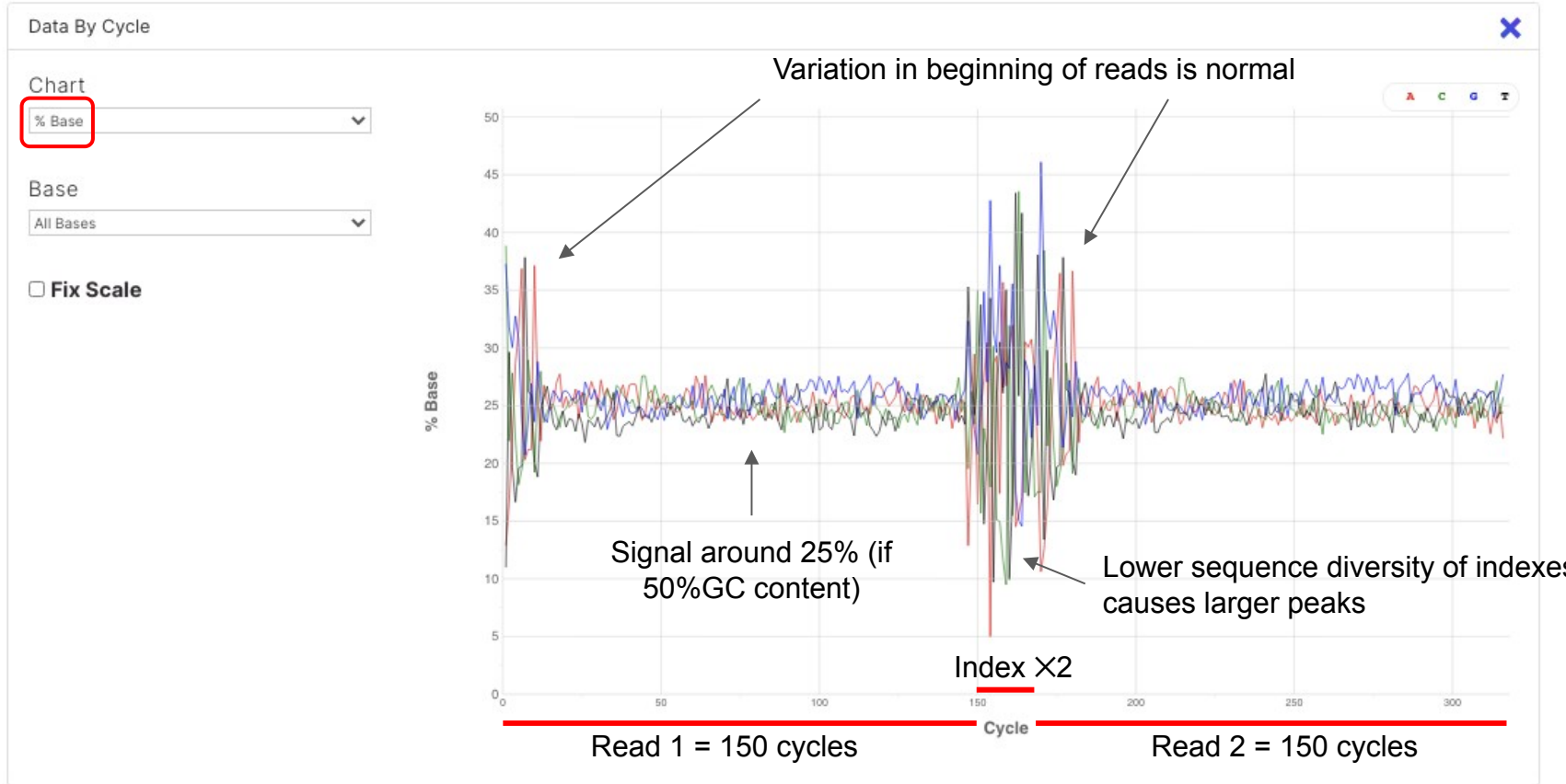
Evaluate level of adapter dimers and fragment length

Data by cycle chart: select **% Base** under 'Chart' -> Adapter dimers? Short fragments?

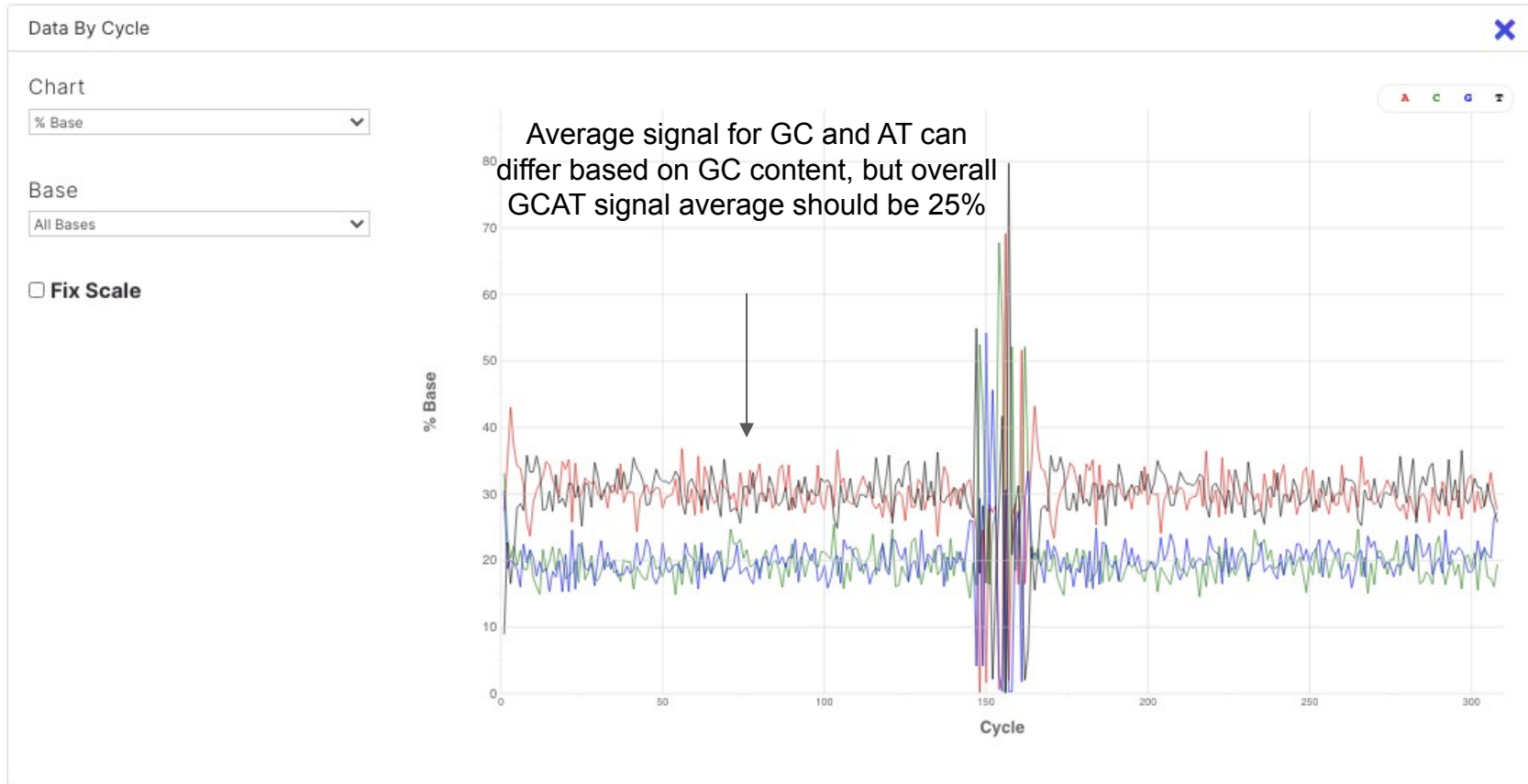


Evaluate level of adapter dimers and fragment length

Data by cycle chart: select **% Base** under 'Chart' -> Adapter dimers? Short fragments?



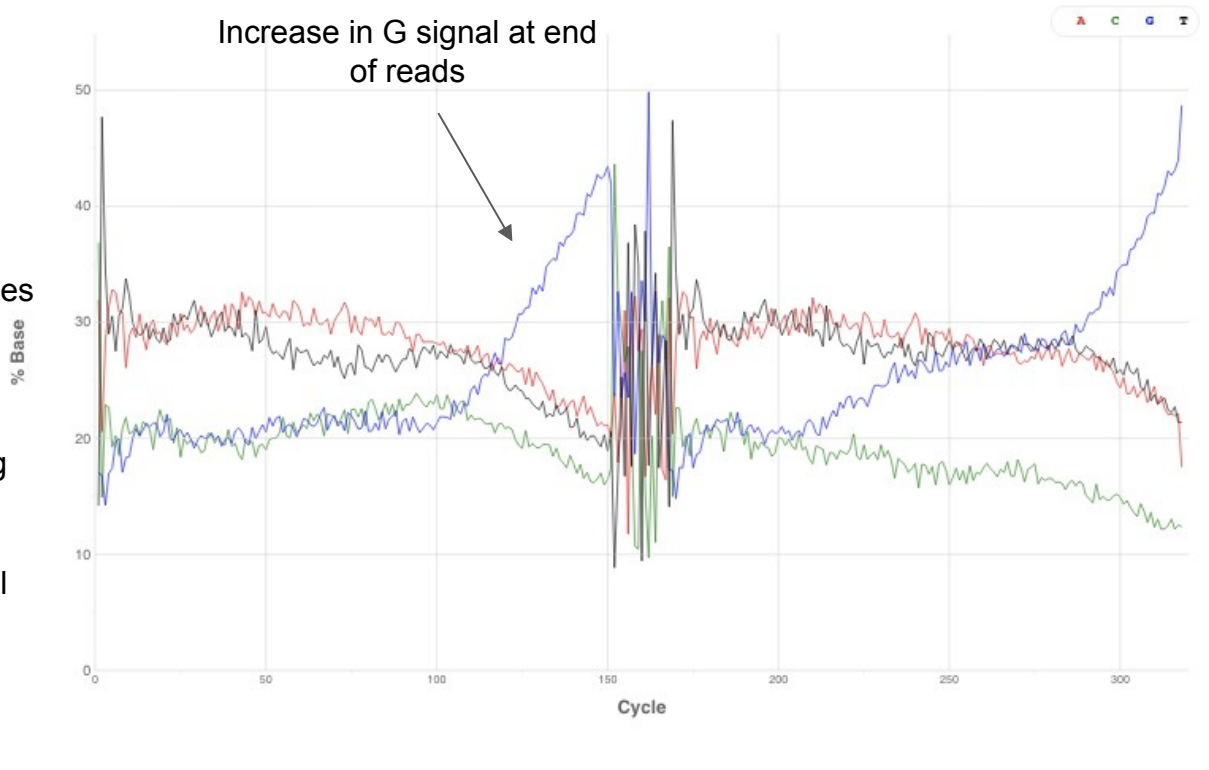
Evaluate level of adapter dimers and fragment length



Evaluate level of adapter dimers and fragment length

Issue 1 - short reads: Increase in G signal at end of reads

- G is the “dark base” is iSeq , MiniSeq, NextSeq and NovaSeq chemistry.
- Continuous increase in G indicates no bases detected for subsequent cycles
 - Inserts shorter than 150 bp?
 - Issue with elongation of strands during clustering?
- If insert size was indeed shorter than usual (check lib prep QC), reduce fragmentation time for future!



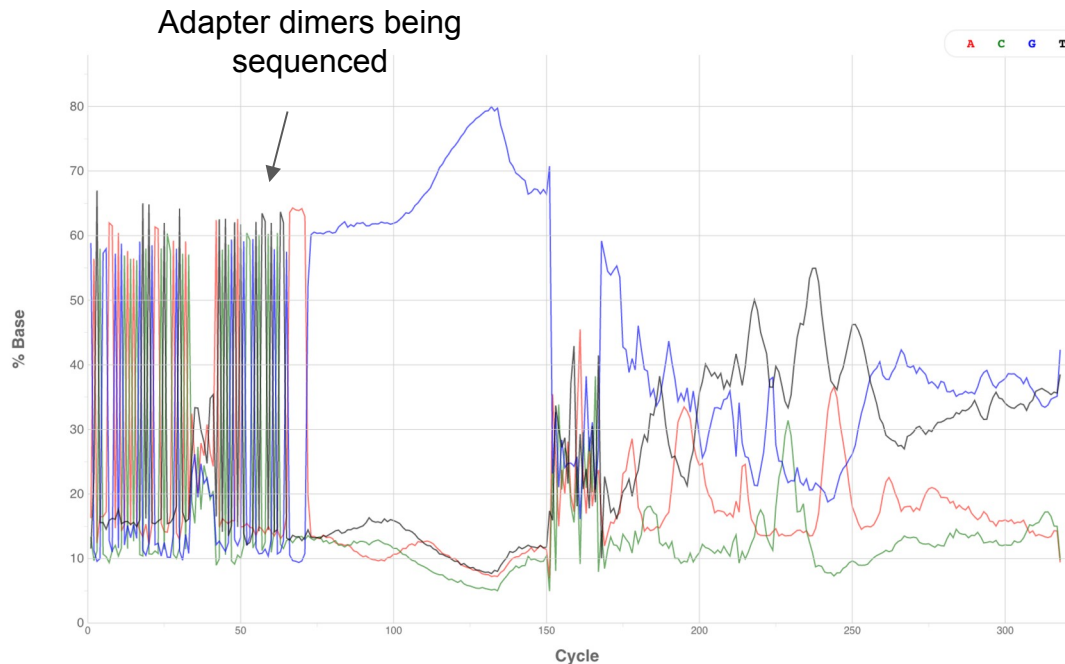
Note: MiSeq and HiSeq would call an A in case nothing is detected.



Evaluate level of adapter dimers and fragment length

Issue 2 - adapter dimers: Large peaks for first 70 cycles, and then an increase in G signal (iSeq, MiniSeq, NextSeq, NovaSeq)

- Adapter dimers will yield reads with low sequence variation (large peaks) that are about 70 bp long
- Adapter dimers are short fragments and thus get preferentially sequenced, taking up valuable sequencing space
- Crucial to remove ALL adapter dimers during library prep QC !!



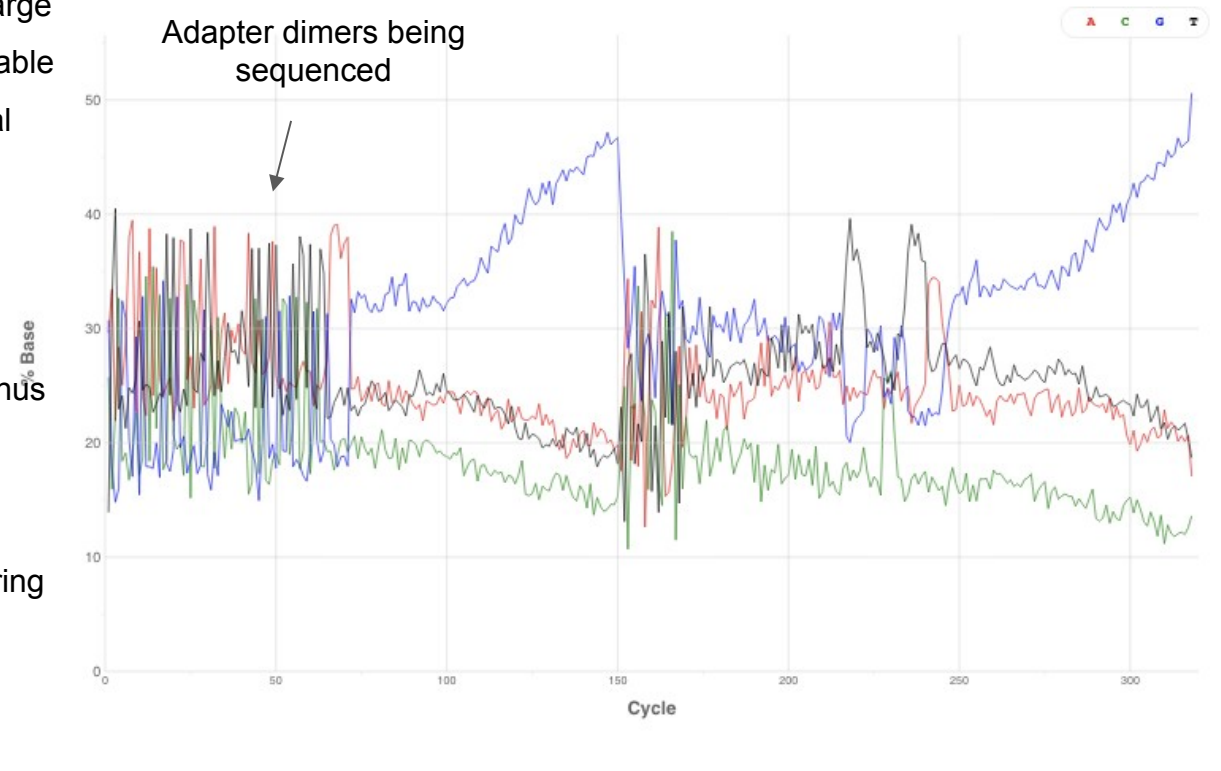
The first ~30 bases are the first portion of the illumina adapter, followed by heterogeneity from the barcode sequences (8-12 bases), followed by the last ~30 bases of the adapter. Then a large spike in the 'dark'/no signal base 'G'.



Evaluate level of adapter dimers and fragment length

Issue 3 - adapter dimers and short reads: Large peaks for first 70 cycles followed by more stable signal and then a steady increase in G signal

- Adapter dimers will yield reads with low sequence variation (large peaks) that are about 70 bp long
- Adapter dimers are short fragments and thus get preferentially sequenced, taking up valuable sequencing space
- Crucial to remove ALL adapter dimers during library prep QC !!



Evaluate error rate based on PhiX

COVID-19 ARTIC_Batch-10

SUMMARY BIOSAMPLES CHARTS **METRICS** INDEXING QC SAMPLE SHEET FILES



Per Read Metrics

READ	CYCLES	YIELD	PROJECTED YIELD	ALIGNED (%)	ERROR RATE (%)	INTENSITY CYCLE 1	%>Q30
Read 1	151	860.56 Mbp	860.56 Mbp	2.72	0.45	259.25	90.79
Read 2 (I)	8	40.16 Mbp	40.16 Mbp	0.00	0.00	210.56	87.55
Read 3 (I)	8	40.16 Mbp	40.16 Mbp	0.00	0.00	376.44	74.09
Read 4	151	860.56 Mbp	860.56 Mbp	2.43	1.09	390.75	87.34
Non-index Reads Total	302	1.72 Gbp	1.72 Gbp	2.58	0.77	325.00	89.06
Total	318	1.80 Gbp	1.80 Gbp	2.58	0.77	309.25	88.69

Evaluate sequencing data yield

COVID-19 ARTIC_Batch-10

SUMMARY BIOSAMPLES CHARTS **METRICS** INDEXING QC SAMPLE SHEET FILES



Scroll down to 'Per Lane Metrics'

Number of paired end reads that were generated

Per Lane Metrics

LANE	STATUS	READ	CLUSTER PF(%)	%≥ Q30	YIELD	ERROR RATE(%)	READS PF	DENSITY	TILES	LEGACY PHAS / PREPHAS(%)	COMMENTS	INTENSITY
☐ 1	<u>QC Passed</u>	Read 1	74.62±1.18	90.79	0.86 Gbp	<u>0.45 ±0.04</u>	<u>5,737,044</u>	326 ±0	16	0.365 / 0.227		259±8
		Read 2 (I)		87.55	0.04 Gbp	<u>0.00 ±0.00</u>				0.000 / 0.000		211±6
		Read 3 (I)		74.09	0.04 Gbp	<u>0.00 ±0.00</u>				0.000 / 0.000		376±18
		Read 4		87.34	0.86 Gbp	<u>1.09 ±0.06</u>				0.269 / 0.127		391±19

Evaluate sequencing metrics

On the 'Metrics' page (if using PhiX)

- Per Read Metrics:
 - a. **% Aligned** (PhiX): should be close to % of PhiX that you spiked in.
 - i. If it's higher, PhiX outcompeted your library, meaning you put less library in than you thought (double check your pooling and quantification calculations).
 - ii. If it's lower than expected, small fragments like adapter dimers might have outcompeted PhiX AND/OR you underestimated the quantity of material in your sample (added way more than you thought, double check your quantification data).
 - b. **Error rate %** (PhiX): should be less than 1%.
- Per Lane Metrics:
 - a. **Reads PF**: number of passed filter reads that were generated in this run. This should be close to (or over) the minimum number of guaranteed reads that your sequencer is advertised to generate. If lower, you probably under- or overclustered. Check your quantification calculations and adjust the loading concentration if needed.

Evaluate demultiplexing

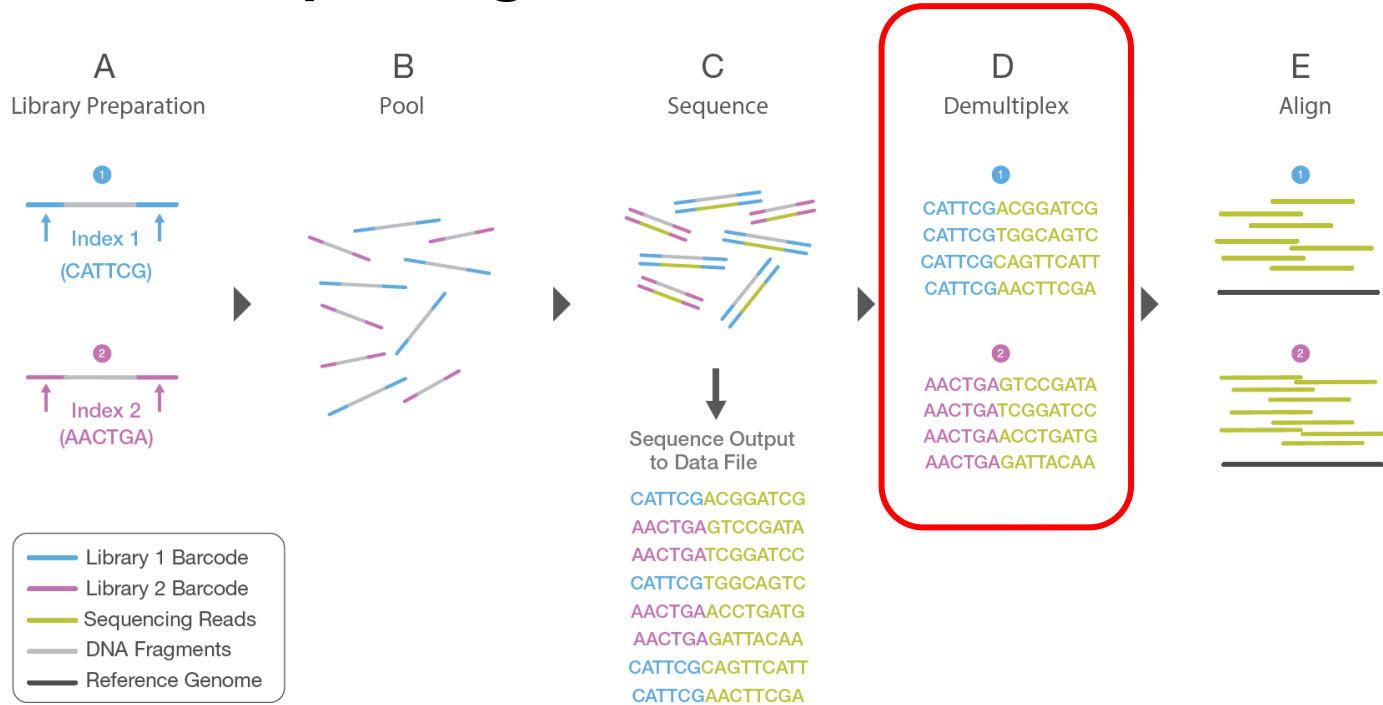


Figure 5: Library Multiplexing Overview.

- Two distinct libraries are attached to unique index sequences. Index sequences are attached during library preparation.
- Libraries are pooled together and loaded into the same flow cell lane.
- Libraries are sequenced together during a single instrument run. All sequences are exported to a single output file.
- A demultiplexing algorithm sorts the reads into different files according to their indexes.
- Each set of reads is aligned to the appropriate reference sequence.

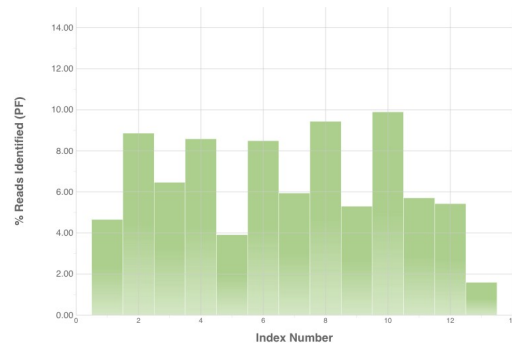
Evaluate demultiplexing

SUMMARY BIOSAMPLES CHARTS METRICS INDEXING QC SAMPLE SHEET FILES

TOTAL READS	PF READS	% READS IDENTIFIED (PF)	% READS UNDETERMINED	CV	MIN	MAX
15,376,000	11,474,088	84.2893	15.7107	0.3772	1.5917	9.8988

INDEX	SAMPLE ID	LIBRARY NAME	INDEX 1 (I7)	INDEX 2 (I5)	% READS IDENTIFIED (PF)
1	COVID_0125	n/a	CGATCGAT	TCGAGTGA	4.6542
2	COVID_0126	n/a	GCCTTAAC	CGCTTAAC	8.8601
3	COVID_0127	n/a	AGTCAGGT	GATAGGCT	6.4675
4	COVID_0128	n/a	CTAGGTTG	ATGGAAGG	8.5830
5	COVID_0129	n/a	CCTCATCT	TAATGCCG	3.9148

If correctly pooled, all samples should have more or less the same # of reads assigned



Evaluate demultiplexing

On the 'Indexing QC' page:

- a. **% read identified** = reads with identified barcodes: should be >80% (the higher the better). If this % is low, you should check to make sure your sample sheet assigned the correct index/barcode sequences for each sample.

- a. **% reads undetermined** = reads without an associated known barcode. If this % is high, you should check to make sure your sample sheet was correct.

- a. **PF reads**: this number details the amount of (unpaired) single end reads. This number is always double than your true amount of paired-end reads shown on the 'Metrics' page.

How to troubleshoot demux in BaseSpace

Locating and interpreting the demux file to troubleshoot failed demultiplexing or missing samples

- A high % of undetermined reads (90%+) and missing samples (samples with no reads assigned) indicate an issue with demultiplexing!
 - Double check the sample sheet to verify if the correct barcodes were assigned to each sample.
 - **Look at the barcodes sequences identified for ‘undetermined reads’ listed in DemuxSummaryF1L1.txt**, if there is barcode combination (paired end) that occurred in a high number of reads, it might be an unidentified sample.
 - In Basespace, go to Summary page of run -> Click on blue “FASTQ Generation of...” under Latest analysis, scroll down to Logs -> click on “View files”.

Info on finding troubleshooting and finding the demux file in BaseSpace [here](#)

-How to edit sample sheet and requeue in Basespace [here](#)

On the summary page of the run, click on *FastQ Generation* under *Latest Analysis*

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SEQUENCE HUB

Manu Vanaersch...

HOME

RUNS

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DEMO DATA

We are currently experiencing unusually high demand on our DRAGEN app launches. Expect delays of upwards of 8 hours while we work with our partners to resolve the issue. We do not recommend to abort and requeue multiple sessions as this can cause further delays. We apologize for the inconvenience.

20211104_COVIDSEQ0.5

SUMMARYBIOSAMPLESCHARTSMETRICSINDEXING QCSAMPLE SHEETFILES

Instrument
MN01689

93.55

Avg
%Q30

Run Status
Complete

Latest Analysis
[FASTQ Generati...](#)

Lane QC Status
QcPassed

Cycles
101 | 10 | 10

Flow Cell Status
QcPassed

Yield
3.23 Gbp

Created
2021-11-04 14:24

File Count/Size
[419 files](#) (2 GB)

Owner
Manu Vanaersch...

Flow Cell ID
000H3MFML

Instrument Type
MiniSeq

File Status
Active

User
Manu Vanaersch...

Run ID
211104_MN0168...

On the summary page of the FASTQ analysis, click on *View Files* next to *Logs*.

We are currently experiencing unusually high demand on our DRAGEN app launches. Expect delays of upwards of 8 hours while we work with our partners to resolve the issue. We do not recommend to abort and requeue multiple sessions as this can cause further delays. We apologize for the inconvenience.

Analysis: FASTQ Generation 2021-11-08 22:11:55Z

Project [20211105_COVIDSEQ0.5x](#)

SUMMARY REPORTS INPUTS FILES



General Info

Name	FASTQ Generation 2021-11-08 22:11:55Z
Application	FASTQ Generation Version: 1.0.0
Date Started	2021-11-08 14:11
Date Completed	2021-11-08 14:21
Duration	9 minutes 20 seconds
Compute Charge	0.00 iCredits
Session Type	Single Node
Size	1.55 GB
Status	Complete
Delivery	None

Logs [View Files](#)

Last checked: 6:18:28 PM UTC

```
2021-11-08 22:16:17.492 [INFO] - Configuring BaseSpace FileSystem driver
2021-11-08 22:16:18.362 [INFO] - Mounting Inputs
2021-11-08 22:16:28.667 [INFO] - Mounting Genomes
2021-11-08 22:16:28.687 [INFO] - Using Genomes from S3
2021-11-08 22:16:37.000 [INFO] - Preparing to download image
2021-11-08 22:16:37.063 [INFO] - Downloading application image (attempt 1).
2021-11-08 22:17:29.649 [INFO] - Finished downloading application image.
2021-11-08 22:17:30.359 [INFO] - Starting application container (attempt 1).
2021-11-08 22:17:31.414 [INFO] - Application container is running.
```

Click on the *DemuxSummaryF1L1.txt* file to download it.

We are currently experiencing unusually high demand on our DRAGEN app launches. Expect delays of upwards of 8 hours while we work with our partners to resolve the issue. We do not recommend to abort and requeue multiple sessions as this can cause further delays. We apologize for the inconvenience.

Analysis: FASTQ Generation 2021-11-08 22:11:55Z

Project [20211105_COVIDSEQ0.5x](#)

SUMMARY REPORTS INPUTS FILES



NAME	CREATED	TYPE	PATH	SIZE
 AdapterTrimming.txt	2021-11-08 1...	txt	AdapterTrimming.txt	6.93 KB
 AppSession.json	2021-11-08 1...	json	AppSession.json	13.20 KB
 CompletedJobInfo.xml	2021-11-08 1...	xml	CompletedJobInfo.xml	4.89 KB
 ConversionStats.xml	2021-11-08 1...	xml	ConversionStats.xml	858.20 KB
 DemultiplexingStats.xml	2021-11-08 1...	xml	DemultiplexingStats.xml	7.28 KB
 DemuxSummaryF1L1.txt	2021-11-08 1...	txt	DemuxSummaryF1L1.txt	31.87 KB

Open the DemuxSummaryF1L1.txt file in a text editor (e.g. Notepad, TextEdit, SublimeText)

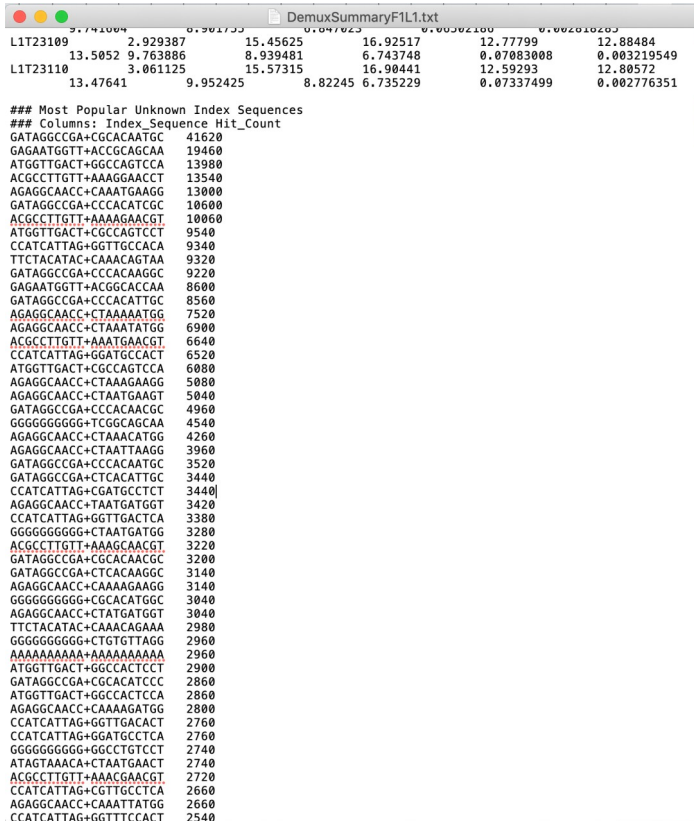
SampleNumber	0	1	2	3	4	5	6	7	8	9
10										
SampleName	None	CT_20_05X	CT_22_05X	CT_24_05X	CT_26_05X					
CT_28_05X	CT_30_05X	CT_32_05X	CT_34_05X	CT_H20_1_05X	CT_H20_2_05X					
L1T11102	6.527649	15.33578	16.65971	12.42023	10.69091					
13.20399	9.823663	8.288112	6.977473	0.06874051						
0.003738155										
L1T11103	6.34731	15.27477	16.70565	12.32653	10.85563					
13.16495	9.838663	8.376467	7.042014	0.06490813	0.003110613					
L1T11104	5.187835	15.2982	16.7218	12.48951	11.92585	13.17055				
9.831048	8.392179	6.911465	0.06820869	0.003347666						
L1T11105	4.812489	15.22733	16.78101	12.58195	12.19453					
13.3017	9.771131	8.401733	6.856949	0.0698997	0.001270904					
L1T11106	4.684128	15.18373	16.88012	12.59622	12.12434					
13.36305	9.785199	8.428674	6.886028	0.06454672						
0.003968892										
L1T11107	4.338824	15.23912	16.79444	12.51882	12.40396					
13.42278	9.902044	8.485727	6.823102	0.06782366						
0.003359686										
L1T11108	3.963507	15.41056	16.76803	12.57677	12.52411					
13.47618	9.825658	8.673107	6.715884	0.06365467						
0.002537728										
L1T11109	3.551315	15.69837	16.90989	12.52709	12.53913					
13.40251	9.786554	8.817552	6.693228	0.07034607						
0.00401374										
L1T11110	3.377234	15.72436	16.91751	12.47396	12.51334					
13.48564	9.916151	8.757728	6.763784	0.06669772						
0.00359956										
L1T12101	3.605445	15.57501	16.79141	12.60352	12.42239					
13.30719	9.830542	8.758068	7.033031	0.07093953						
0.00245324										
L1T12102	3.67481	15.68254	16.8822	12.68154	12.03864	13.23109				
9.848054	8.772027	7.111419	0.07418031	0.003502959						
L1T12103	3.92712	15.7765	16.99797	12.73116	11.23678	13.27387				
10.0461	8.869562	7.069814	0.06781297	0.003328244						
L1T12104	3.711355	15.45256	17.08726	12.5363	12.1697	13.32954				
9.921028	8.803834	6.920935	0.06499891	0.00248415						
L1T12105	3.540319	15.52281	16.91424	12.69377	12.37834					
13.41021	9.845559	8.88296	6.741698	0.06865399	0.001443177					
L1T12106	3.394781	15.42443	16.93253	12.60558	12.51725					
13.49617	9.967749	8.818157	6.766393	0.0728192						
0.004137455										
L1T12107	3.486646	15.48401	16.83296	12.52976	12.52789					
13.54599	9.911765	8.927085	6.67627	0.07451518	0.003113448					
L1T12108	3.392642	15.61235	16.89482	12.39576	12.5793					
13.48223	9.922054	8.958322	6.699647	0.06776138	0.003117855					
L1T12109	3.563795	15.59247	16.81039	12.41517	12.48179					
13.51098	9.933734	8.8684	6.74971	0.07103631	0.002521999					
L1T12110	3.59298	15.67375	16.91117	12.24472	12.42779					
13.5104	9.88376	8.855175	6.82502	0.0721048	0.003116922					
L1T13102	3.918591	15.78206	17.03826	12.54448	11.78177					
13.07634	9.844135	8.868717	7.070753	0.07138215						
0.003507216										
L1T13103	4.021693	15.83637	17.31709	12.6986	10.80409					
13.05793	10.0654	8.892047	7.223732	0.08033788	0.002712708					
L1T13104	3.726185	15.70846	17.16058	12.60147	11.89799					
13.26292	9.766609	8.801566	6.989521	0.07972995						
0.004970179										
L1T13105	4.232028	15.6540	17.07081	12.55077	12.38005					

How it looks when you open the file

Scroll down to the section with header
“### Most Popular Unknown Index
Sequences”

13.20399	9.823663	8.288112	6.977473	0.06874051						
13.16495	9.838663	8.376467	7.042014	0.06490813	0.003110613					
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Inspect the top barcode combinations listed under ###Most Popular Unknown Index Sequences



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### Most Popular Unknown Index Sequences
### Columns: Index_Sequence Hit_Count
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GAGAAATGGTT+ACGGCAGCAA 19460
ATGGTTGACT+GGCCAGTCCA 13980
ACGCCTTGTT+AAAGGAACCT 13540
AGAGGCAACC+CAATGAAGG 13080
GATAGGCCGA+CCACATCCG 10600
ACGCCTTGTT+AAAAGAACGT 10060
ATGGTTGACT+CGCCAGTCTC 9540
CCATCATTAG+GGTTGCACAC 9340
TTCTACATAC+CAAAAGATA 9320
GATAGGCCGA+CCACAAAGGC 9220
GAGAAATGGTT+ACGGCAGCAA 8600
GATAGGCCGA+CCACATCCG 8560
AGAGGCAACC+CTAAATATGG 7520
AGAGGCAACC+CTAAATATGG 6900
ACGCCTTGTT+AAATGAACGT 6640
CCATCATTAG+GGATGCCACT 6520
ATGGTTGACT+CGCCAGTCCA 6080
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AGAGGCAACC+CTAATGAAGT 5040
GATAGGCCGA+CCACAAAGC 4960
GGGGGGGGG+TCGGCAGCAA 4540
AGAGGCAACC+CTAAACATGG 4260
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GATAGGCCGA+CTCACATTCG 3440
CCATCATTAG+CGATGCTCT 3440
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CCATCATTAG+GGTTGACTCA 3380
GGGGGGGGG+CTAATGATGG 3280
ACGCCTTGTT+AAAGCAACGT 3220
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GATAGGCCGA+CTCACAAAGC 3140
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AGAGGCAACC+CTATGATGGT 3040
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AAAAAAAAAA+AAAAAAAAAA 2960
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GATAGGCCGA+CGCACATCC 2860
ATGGTTGACT+GGCCACTCCA 2860
AGAGGCAACC+CAAAAGATGG 2800
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CCATCATTAG+GGATGCCCTCA 2760
GGGGGGGGG+GGCCTGTCTC 2740
ATAGTAACA+CTAATGAAGT 2740
ACGCCTTGTT+AAAGCAACGT 2720
CCATCATTAG+CGTTGCTCTCA 2660
AGAGGCAACC+CAAAATATGG 2660
CCATCATTAG+GGTTTCACT 2540
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Hit_Count = the number of reads that have the listed barcode combination (NNNNNNNN....+NNNNNNNN....)

Focus on barcode combinations that have a high Hit_Count!

Common issues:

- Barcode/index sequence in sample sheet was (not) reverse complemented
- Too many index cycles performed -> extra nucleotides were added to the barcodes/indexes. Identify which nucleotides in this list and update sample sheet for requeuing fastq analysis.

When the samplesheet is corrected/updated -> requeue for demux

How to requeue demux in BaseSpace

On the *Summary* page of the run, click on the hourglass, select *Requeue* and then select *Sample Sheet*

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SEQUENCE HUB

Manu Vanaersch... 



 HOME RUNS PROJECTS ANALYSES BIOSAMPLES APPS DEMO DATA   

We are currently experiencing unusually high demand on our DRAGEN app launches. Expect delays of upwards of 8 hours while we work with our partners to resolve the issue. We do not recommend to abort and requeue multiple sessions as this can cause further delays. We apologize for the inconvenience.

SUMMARY BIOSAMPLES CHARTS METRICS INDEXING QC SAMPLE SHEET FILES

STATUS

CANCEL >

CHANGE >

REQUEUE >

UNLOCK >

BIOSAMPLE

FASTQ GENERATION

LIBRARY

POOL

SAMPLE SHEET

Run Status
Complete

Latest Analysis
FASTQ Generati...

Cycles
101 | 10 | 20

Yield
3.23 Gbp

Cell Status
Completed

Created
2021-11-04 14:24

Instrument Type
MiniSeq

File Count/Size
419 files (2 GB)

File Status
Active

Owner
Manu Vanaersch...

User
Manu Vanaersch...

Flow Cell ID
000H3MFML

Run ID
211104_MN0168...

Click with your cursor in the black part of the screen and select everything (ctrl+A or cmd+A)

illumina

SEQUENCE HUB

Manu Vanaersc...  



HOME

RUNS

PROJECTS

ANALYSES

BIOSAMPLES

APPS

DEMO DATA







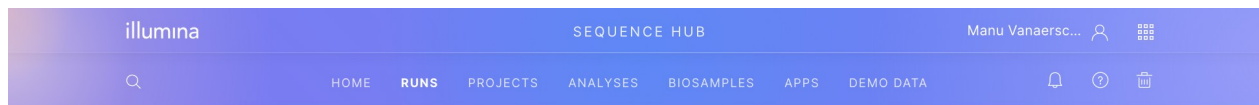
Please enter a new sample sheet, or load the original to edit it. You may generate a new Sample Sheet using [Illumina Experiment Manager](#).

[Load Original](#) **QUEUE ANALYSIS**

Sample sheet is valid.

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7 Application,FASTQ Only,,,,,,,,
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24 COVIDSEQ,COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_28_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_28_05X,CT_28_05X,Lusajo_Mwakibete,ATGGTTGACT,ATGGTTGACT,AGGACAGGCC,AGGACAGGCC
25 COVIDSEQ,COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_30_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_30_05X,CT_30_05X,Lusajo_Mwakibete,TATTGGCTCT,TATTGGCTCT,CCTAACACAG,CCTAACACAG
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28 COVIDSEQ,COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_H20_1_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_H20_1_05X,CT_H20_1_05X,Lusajo_Mwakibete,AACCATAGAA,AACCATAGAA,CCATCTCGCG,CCATCTCGCG
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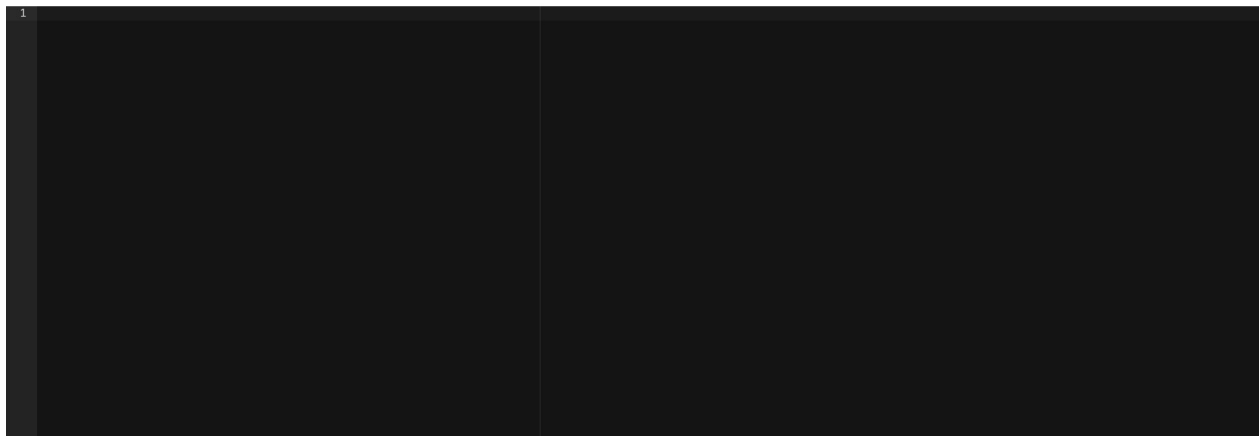
Click delete to delete the content of the sample sheet



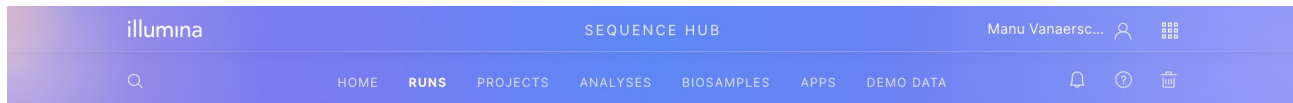
Please enter a new sample sheet, or load the original to edit it. You may generate a new Sample Sheet using [Illumina Experiment Manager](#).

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QUEUE ANALYSIS



Open your corrected sample sheet (saved as a .csv file) in a text editor such as Notepad or TextEdit, select all of its content (ctrl+A or cmd+A) and copy it into the black screen.



Please enter a new sample sheet, or load the original to edit it. You may generate a new Sample Sheet using [Illumina Experiment Manager](#).

[Load Original](#)

[QUEUE ANALYSIS](#)

Sample sheet is valid.

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3 Investigator Name,ManuVanaerschat,,,,,,,,
4 Experiment Name,20211021_RR153_RR188e_demorun,,,,,,,,
5 Date,10/21/21,,,,,,,,
6 Workflow,GenerateFASTQ,,,,,,,,
7 Application,FASTQ Only,,,,,,,,
8 Assay,Nextera XT,,,,,,,,
9 Description,,,,,,,,
10 Chemistry,Amplicon,,,,,,,,
11 ,,,,,,,,,
12 [Reads],,,,,,,,,
13 101,,,,,,,,
14 ,,,,,,,,,
15 [Settings],,,,,,,,,
16 ReverseComplement,0,,,,,,,,
17 ,,,,,,,,,
18 [Data],,,,,,,,,
19 Study_ID,Study_Description,BioSample_ID,BioSample_Description,Sample_ID,Sample_Name,Sample_Owner,Index_ID,Index,Index2_ID,Index2_Organism
20 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_20_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_20_05X,CT_20_05X,Lusajo_Mwakibete,GAGAAATGGTT,GAGAAATGGTT,TTGCTGCCGA,TTGCTGCCGA
21 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_22_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_22_05X,CT_22_05X,Lusajo_Mwakibete,AGAGGCAACC,AGAGGCAACC,CCATCATTAG,CCATCATTAG
22 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_24_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_24_05X,CT_24_05X,Lusajo_Mwakibete,CCATCATTAG,CCATCATTAG,AGAGGCAACC,AGAGGCAACC
23 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_26_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_26_05X,CT_26_05X,Lusajo_Mwakibete,GATAGGCCGA,GATAGGCCGA,GCCATGTGGC,GCCATGTGGC
24 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_28_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_28_05X,CT_28_05X,Lusajo_Mwakibete,ATGTTGACT,ATGTTGACT,AGGACAGGCC,AGGACAGGCC
25 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_30_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_30_05X,CT_30_05X,Lusajo_Mwakibete,TATTGCGCTC,TATTGCGCTC,CTTAACACAG,CTTAACACAG
26 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_32_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_32_05X,CT_32_05X,Lusajo_Mwakibete,AGCCCTGTGT,AGCCCTGTGT,AGCTTCGTTA,AGCTTCGTTA
27 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_34_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_34_05X,CT_34_05X,Lusajo_Mwakibete,TTCTACATAC,TTCTACATAC,TTACAGTTAG,TTACAGTTAG
28 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_H20_1_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_H20_1_05X,CT_H20_1_05X,Lusajo_Mwakibete,AACCATAGAA,AACCATAGAA,CCATCTCGCG,CCATCTCGCG
29 COVIDSEQ_COVIDSEQ_exploration_comparison_with_ARTIC_illumina,CT_H20_2_0_5X,Nasal_Swab_Extractions_COVID_Positive_Patients,CT_H20_2_05X,CT_H20_2_05X,Lusajo_Mwakibete,GGTTGCGAGG,GGTTGCGAGG,TTGCTCTATT,TTGCTCTATT
```

A green banner saying “Sample sheet is valid” will re-appear automatically if the copied sample sheet is valid. If not, make sure there are no columns or data fields missing in your sample sheet and that you indeed copied the full contents of the sample sheet into the field.

If sample sheet was valid, click on the blue button for *Queue Analysis*

What if my sequencer is not connected to the internet?

(a guide to SAV)

If your sequencer is not connected to the internet...

- Your run cannot be uploaded / viewed on Basespace

but...

- You can use the Illumina app: **Sequencing Analysis Viewer (SAV)**

Illumina Sequencing Analysis Viewer (SAV)

Illumina has a software app ([Sequencing Analysis Viewer](#))

You can download the app and use it to visualize the metrics/data included in the run folder, just as you would use BaseSpace

There are a few short training videos on the [SAV website](#) that we recommend

SAV user guide can be found [here](#)

1. Retrieve your run files from the Illumina sequencer

You will need to transfer the following files from the sequencer to your computer where you are using SAV

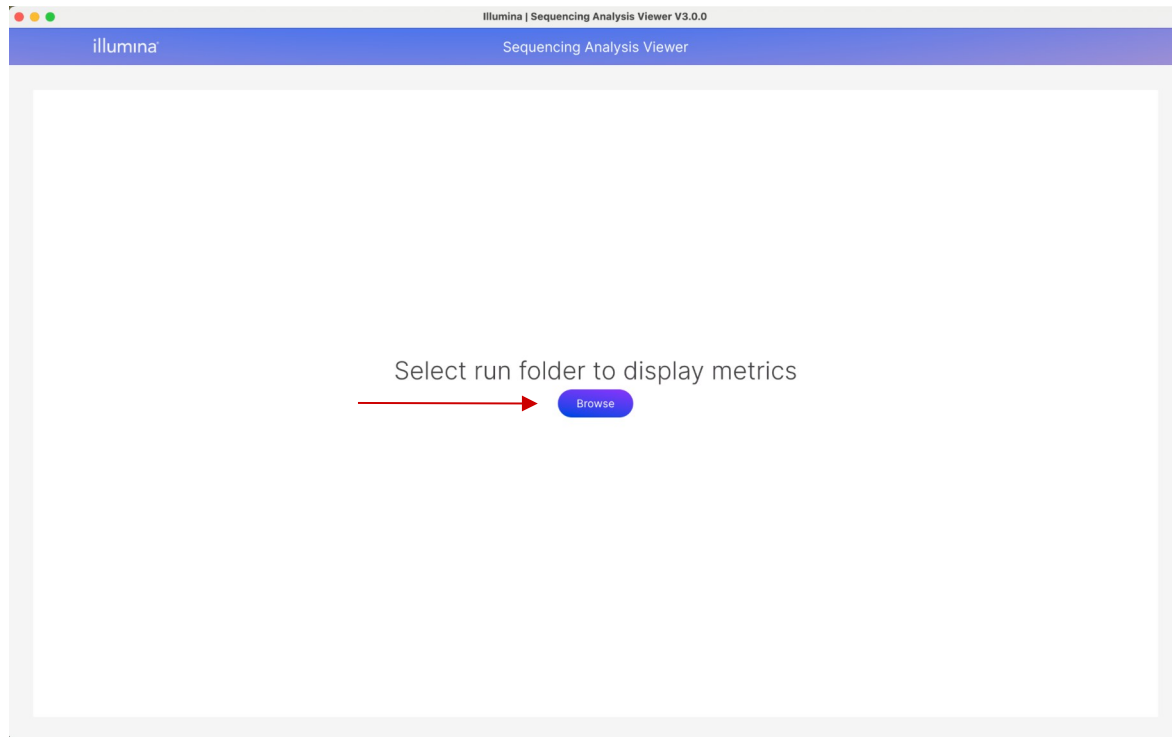
▼ Input Files

SAV uses the following files from the run output folder as input:

- RunInfo.xml
- RunParameters.xml
- InterOp files. InterOp files provide the following information:
 - Extraction metrics
 - Quality metrics
 - Error metrics
 - Tile metrics
 - Extended tile metrics
 - Corrected intensity metrics
 - Image metrics
 - Index metrics
 - Empirical phasing metrics
- **[Optional]** Thumbnails

2. Open run in Sequence Analysis Viewer

After retrieving your sequencing run files and opening Sequence Analysis Viewer, click on 'Browse' and select the appropriate Run folder to view its metrics.



3. Follow the earlier slides in this deck for step by step QC

The process of reviewing QC in SAV is the same as the process you would follow in Basespace

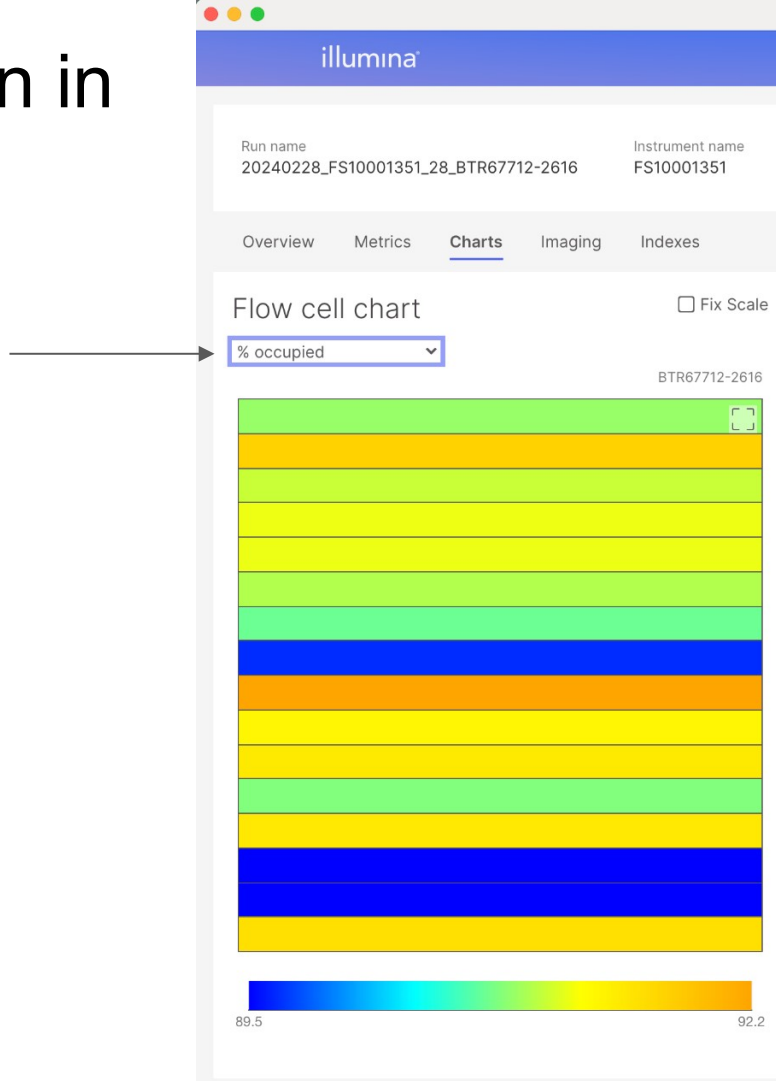
How to QC a sequencing run in Sequence Analysis Viewer

For iSeq runs (patterned flow cell):

In the Flow Cell Chart, click on the arrow to display the '% Occupied' data instead of 'Intensity'.

Determine if the optimal amount of library was loaded onto the flow cell. Are there any signs of over/under clustering?

The % Occupied ranges here from 71 to 76. This indicates minor underclustering. This does not necessarily affect sequence quality, but it does reduce the maximum yield of data that can be obtained from a sequencing run. For the next sequencing run the researchers can load more material on the flow cell (provided that the same quantification protocol is being used) to generate more sequencing data from a single run .



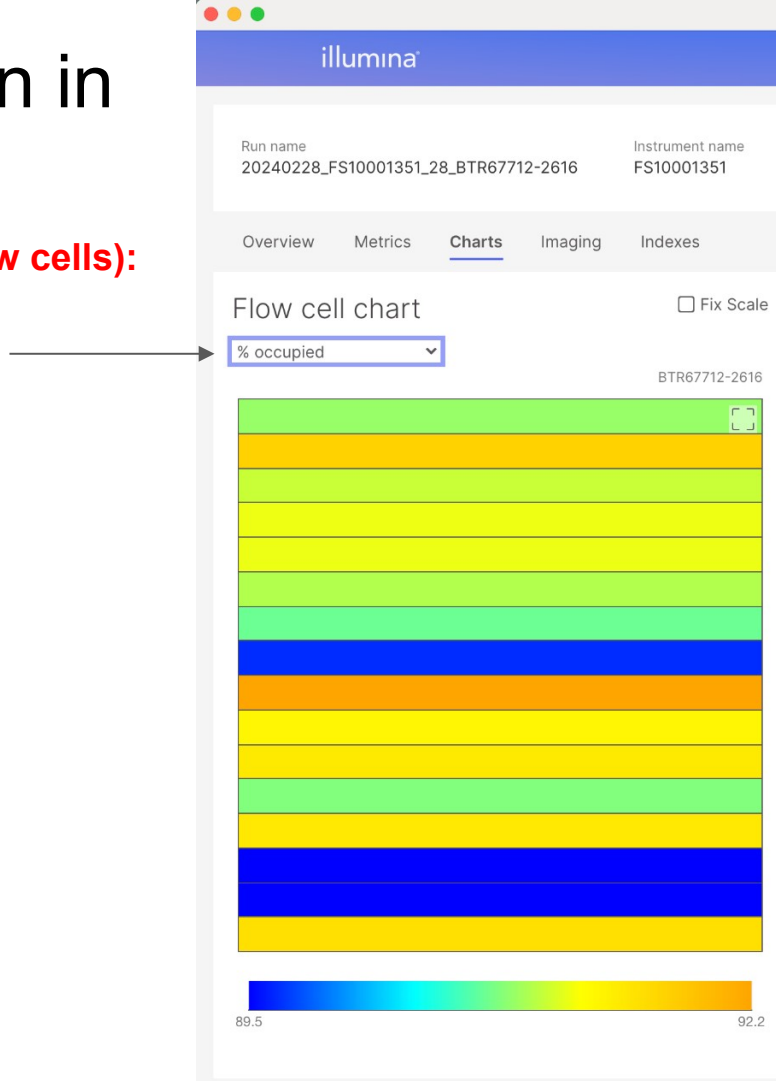
How to QC a sequencing run in Sequence Analysis Viewer

For MiSeq / MiniSeq runs (unpatterned flow cells):

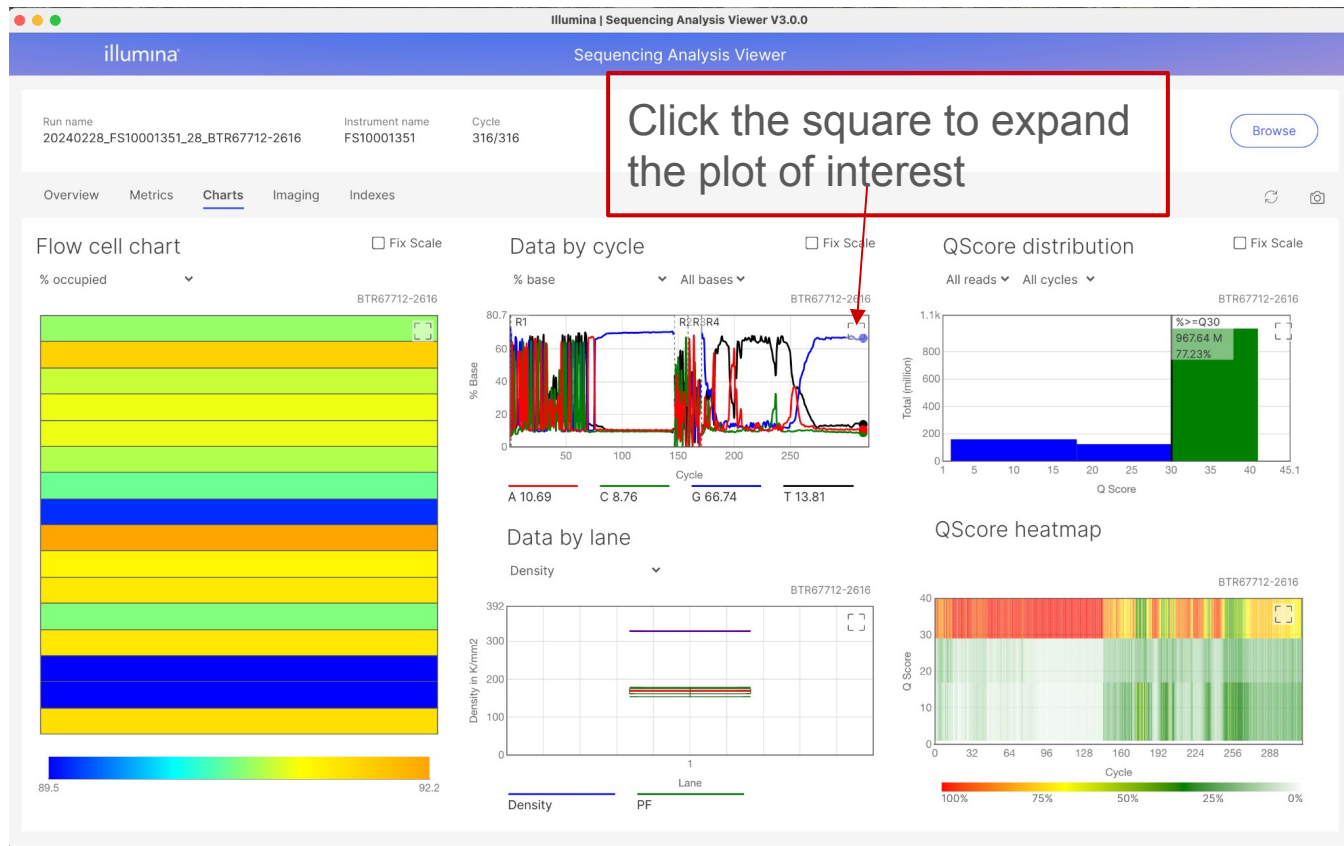
In the Flow Cell Chart, click on the arrow to display the 'Density' data instead of 'Intensity'.

Determine if the optimal amount of library was loaded onto the flow cell. Are there any signs of over/under clustering?

The optimal Density ranges are dependent on the flow cell / sequencer used. View [Illumina's recommendations for cluster density](#) here.

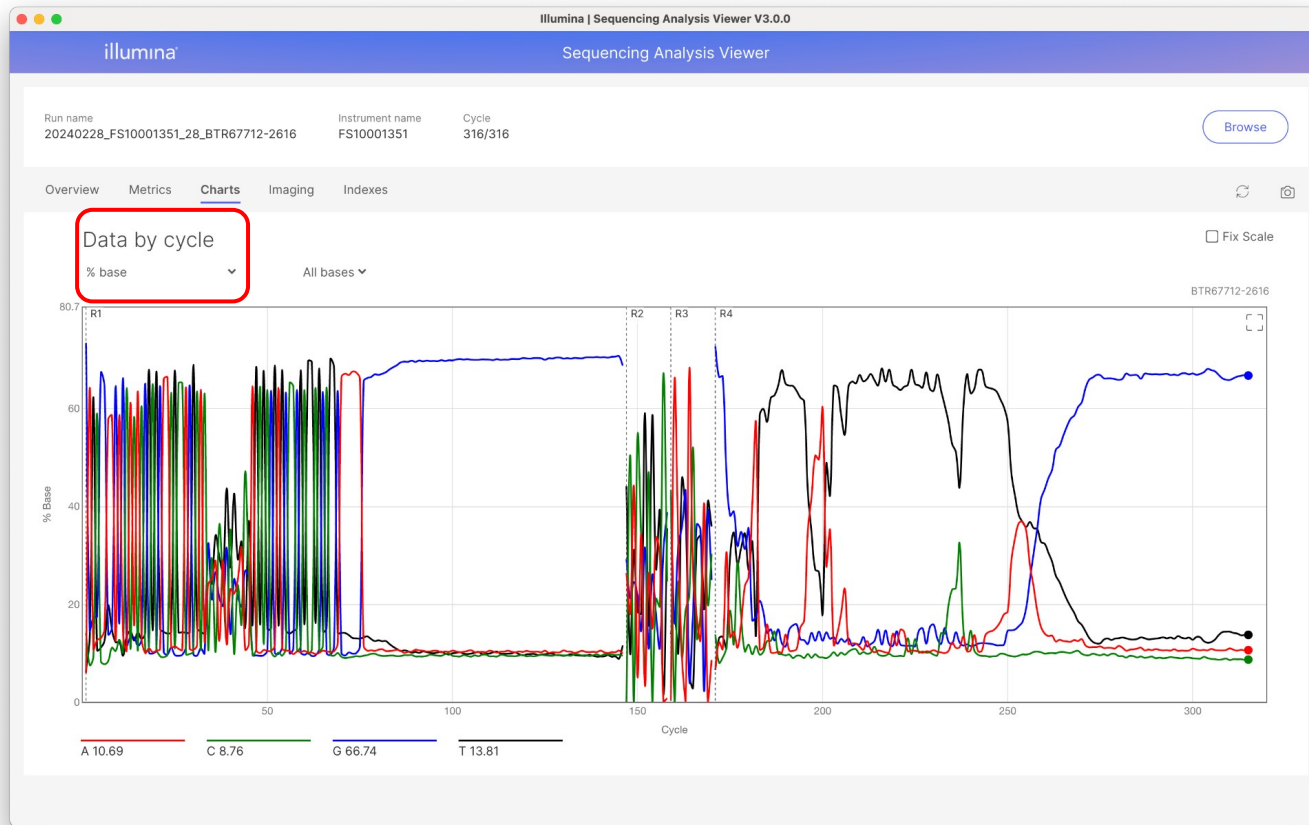


How to QC a sequencing run in Sequence Analysis Viewer



How to QC a sequencing run in Sequence Analysis Viewer

In the 'Data by Cycle' graph, select '% Base' to determine if you had any adapter dimers, short fragments or potentially other issues.



How to QC a sequencing run in Sequence Analysis Viewer

Click on the 'Summary' tab to view how many reads were generated, and if the % aligned and the error rate are within the expected range.

Analysis Imaging **Summary** Indexing

Cycle Status

Extracted: 316 Called: 316 Scored: 316 Error Rated: 315

Run Summary

Level	Yield Total (G)	Projected Total Yield (G)	Aligned (%)	Error Rate (%)	Intensity Cycle 1	% >= Q30
Read 1	0.73	0.73	1.19	0.31	220	95.80
Read 2 (I)	0.06	0.06	0.00	NaN	193	89.87
Read 3 (I)	0.06	0.06	0.00	NaN	257	70.38
Read 4	0.73	0.73	1.16	0.44	241	94.26
Non-Indexed Total	1.46	1.46	1.17	0.37	230	95.03
Total	1.57	1.57	1.17	0.37	228	93.98

% aligned should be similar to %PhiX that was spiked in

Error rate (%) should be <1.0

Cluster Count PF (M) shows how many paired-end passed filter reads were generated (in this case 5.03 million PF reads)

Read 1

Lane	Tiles	Density (K/mm2)	Cluster PF (%)	Legacy Phasing/Prephasing Rate	Phasing slope/offset	Prephasing slope/offset	Cluster Count Raw (M)	Cluster Count PF (M)	% >= Q30	Yield (G)	Cycles Err Rated	Aligned (%)	Error Rate (%)	Error F 35 cycle
1	16	327 ± 0	65.44 ± 0.99	0.161 / 0.053	0.128 / 4.896	0.140 / 1.598	7.69	5.03	95.80	0.73	145	1.19 ± 0.03	0.31 ± 0.04	0.15 ± 0

How to QC a sequencing run in Sequence Analysis Viewer

Click on the 'Indexing' tab to view metrics on demultiplexing.

Analysis Imaging Summary **Indexing**

Lane All



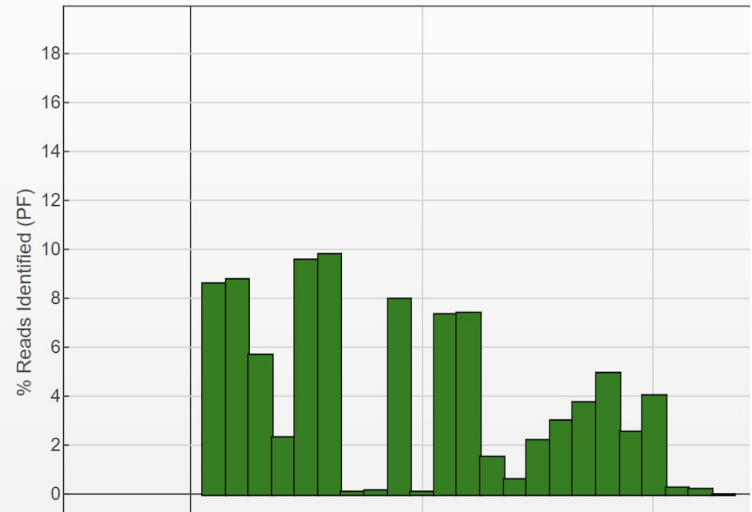
Reads mapped to Index Id

Total Reads	PF Reads	% Reads Identified (PF)	CV	Min	Max
15376000	10062394	90.9343	0.8843	0.0003	9.8336

% Undetermined reads = 100 – '%Reads Identified (PF)'
(= 9.07% in this example)

Check if all samples are equally represented
(and none are missing)

Index Number	Sample Id	Project	Index 1 (I7)	Index 2 (I5)	% Reads Identified (PF)
1	1	default	AGTGCATTGCGAG	GTTTCGAGATTGG	8.5906
2	7	default	ACGAATCGACCA	CTCAAGTCGATG	8.8083
3	8	default	ACGATACGGCAA	TCACATGACGCT	5.7
4	9	default	GAGTTCTTCGGT	TCACTCTCGATC	2.3061
5	12	default	TCACATTCGTCC	CAAGACTGGTTC	9.5821
6	15	default	CTCATGCATCTC	GAGTTAGCAGAC	9.8336
7	20	default	GTTTCAGATGCTG	GTTTCGCCATATC	0.0889
8	21	default	CAAGGCACTGTT	ACGACTGCATCT	0.1773
9	23	default	CTCATGGTCTCT	TCACTAATCGCC	7.9745
10	24	default	ACGAGTAGCTGT	GTTCTACCAAG	0.0661
11	30	default	GTTCTAAGGAGG	CAAGTCCGGTTA	7.375
12	31	default	GAGTGCCTGTAA	GAGTAGTCCAGT	7.3978
13	6	default	GAGTACTCTGCT	GAGTCACGATGT	1.5026
14	10	default	CTCACGTGTCTAT	GTTTCACAGACAC	0.6118
15	13	default	TCACAGGTTTCT	AGTGTGTGCGTA	2.1866
16	14	default	CAAGTCTTCTCG	ACGAACCTCTGA	2.9972
17	19	default	GAGTCTCGATTTC	CTCAACCTCTCA	3.7753
18	25	default	TCACCGACCATT	CAAGCACAACCT	4.9569
19	26	default	CTCATCTACAGC	CAAGCCCTCTAT	0.5276



Run review complete!

Current Activity: 8-I. Sequencing Run QC for Illumina - Activity

Next Slides: DL1_Bioinformatic Basics and BLAST – Slides

Next Activity: DL1_BLAST - Activity